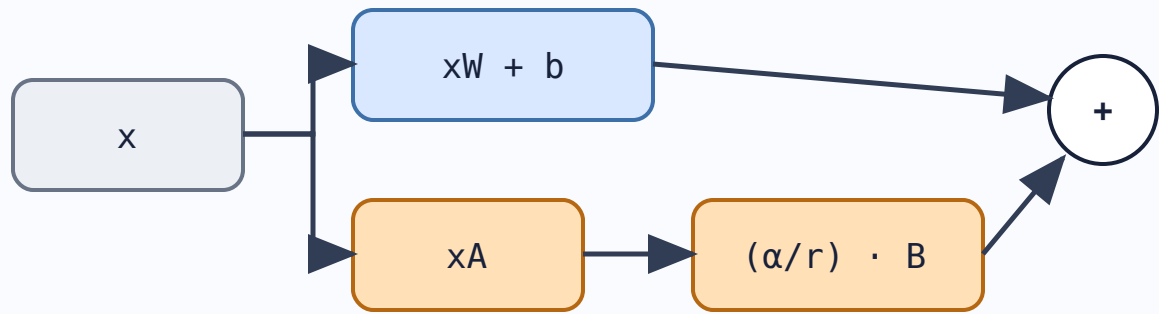


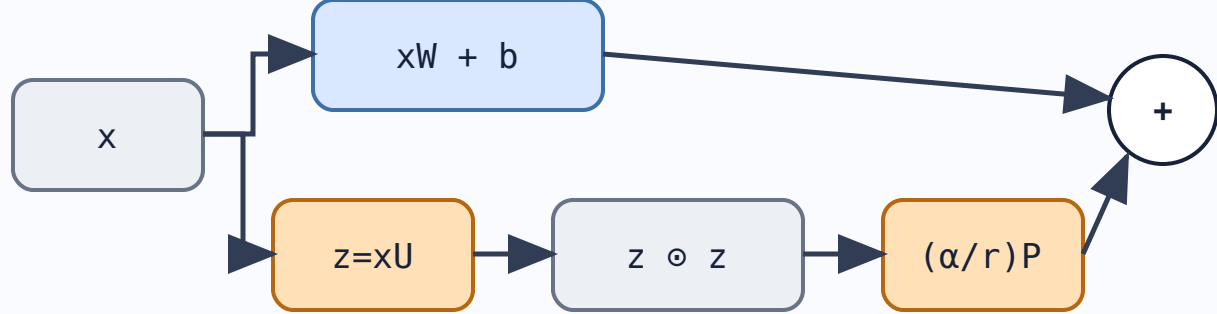
Where the adaptation is applied

LoRA — linear low-rank weight correction



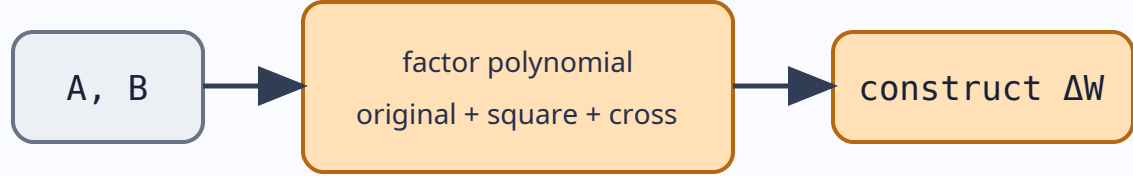
$\Delta y = (\alpha/r)(xA)B$; constant ΔW can be merged.

Symmetric Quadratic — activation-space correction



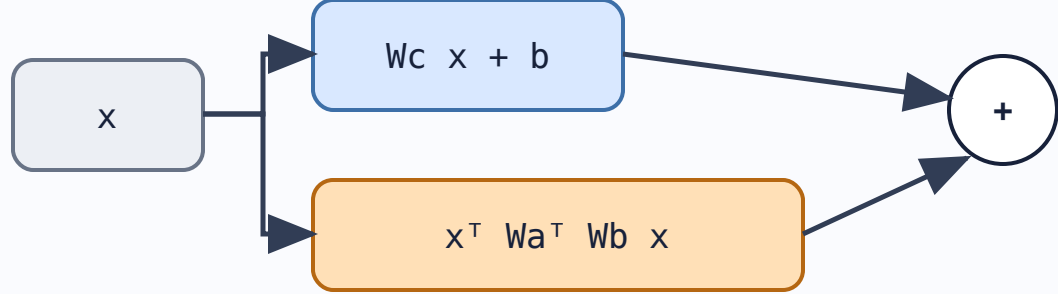
$\Delta y = (\alpha/r)((xU) \odot (xU))P$; input-dependent Jacobian.

PERA — polynomial expansion in parameter-factor space



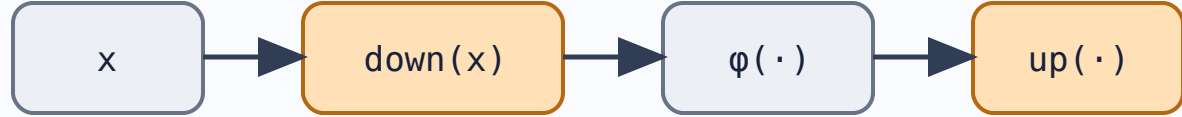
The polynomial is built from trainable factors, not from the current x .
Related motivation; not the adapter implemented in this repository.

QuadraNet V2 — quadratic form added to a pretrained term



Closely related quadratic-form precedent; sharing and placement differ.

Nonlinear bottleneck adapter — general added module



A nonlinear bottleneck allows a general activation choice; it does not impose the shared symmetric form $Q_j = U \text{diag}(p_j) U^T$.

 frozen  trainable  fixed operation